

PAPER_878: SCm Gaussian Activation B-Field Suppression

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Abstract

SCm Gaussian activation function comparing linear $\exp(-B/B_{\text{crit}})$ vs Gaussian $\exp[-(B/B_{\text{crit}})^2]$ magnetic field suppression in the SCm manifold. Gaussian form matches BCS gap behavior with flatter response at low B (preserving coherence) and sharper cutoff near B_{crit} . At $B=0.5 \cdot B_{\text{crit}}$: linear yields 0.607, Gaussian yields 0.779 — 28% higher coherence retention.

1. Core Equations

$$\begin{aligned}\beta(B) &= \exp(-B/B_{\text{crit}}) && [\text{linear}] \\ A_{\text{SCm}}(B) &= \exp[-(B/B_{\text{crit}})^2] && [\text{Gaussian}] \\ \alpha(B) &= w \cdot \beta + (1-w) \cdot A_{\text{SCm}} && [\text{blended}]\end{aligned}$$

2. Parameters

Parameter	Default	Description
B	1e12 T	Applied magnetic field strength
B_crit	4.4e13 T	Critical magnetic field (magnetar scale)
w_blend	0.5	Blending weight between linear and Gaussian

3. UQFF Integration

This calculator integrates `scm_activation_function.py` from Session 204 into the CP4 pipeline as class #462.

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